

Second Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75**Time: 3 hrs.**

SET 'A'

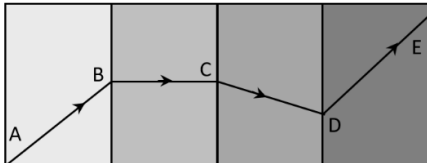
Pass Marks: 30

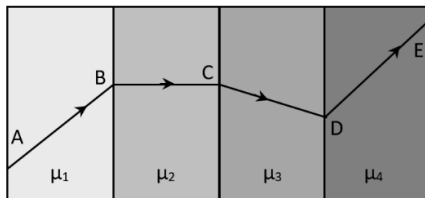
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. Determine the dimensions of ‘a’ in the equation: $\left(P + \frac{a}{v^2}\right)(V - b) = RT$.
a. $[M^1 L^2 T^{-2}]$
b. $[M^1 L^5 T^{-2}]$
c. $[M^1 L^5 T^{-3}]$
d. $[M^{-1} L^2 T^{-2}]$
2. A force of 100 N has a Y- component 50 N. Then X-component of force will be
a. 50 N
b. 86.5 N
c. 70 N
d. 77 N
3. Which two angles of projection gives same horizontal range?
a. $20^\circ, 40^\circ$
b. $30^\circ, 70^\circ$
c. $40^\circ, 45^\circ$
d. $18^\circ, 72^\circ$
4. A body is dropped from the top of a tower. The distance covered in the last second of downward motion is 45m. The height of the tower is:
a. 45m
b. 80m
c. 125 m
d. 180m
5. Rocket propulsion works on the principle of conservation of:
a. linear momentum
b. angular momentum
c. mass
d. torque
6. Two thermometer one Celsius and other Fahrenheit are put in hot bath. The reading on Fahrenheit thermometer is three times the reading on Celsius thermometer. The temperature of hot bath is:
a. 100°C
b. $80/3^\circ\text{C}$
c. 80°C
d. 40°C
7. A ray of light passes through four different parallel mediums with refractive index μ_1, μ_2, μ_3 and μ_4 and if the emergent ray DE is parallel to incident ray AB. Then




- a. $\mu_1 = \mu_2$ b. $\mu_2 = \mu_3$ c. $\mu_3 = \mu_4$ d. $\mu_4 = \mu_1$

8. A man looks his face on a mirror and observes larger image. The mirror must be
 - a. plane mirror
 - b. concave mirror
 - c. convex mirror
 - d. all of the above
9. When a body is charged by induction, then it
 - a. losses some of part of charge
 - b. losses whole charge
 - c. do not lose any charge
 - d. none of the above
10. When 10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere becomes :
 - a. $16 \mu\text{C}$
 - b. $-16 \mu\text{C}$
 - c. $32 \mu\text{C}$
 - d. $-32 \mu\text{C}$
11. The electric field at 20 cm from the centre of uniformly charged hollow metal sphere of radius 10 cm is E. Then, at a distance 5 cm from the centre will be
 - a. $16 E$
 - b. $4 E$
 - c. $2 E$
 - d. 0

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a) A student writes the formula of centripetal force as $F = \frac{mv^2}{r}$, where ‘m’ is the mass, ‘v’ is the velocity and ‘r’ is the radius of the circular path. Is the equation dimensionally correct? [2]
 b) Convert 10^{-7} J into erg using dimensional analysis. [2]
 c) State the principle of homogeneity and give one example. [1]
13. Two vectors $\vec{A}$ and $\vec{B}$ are acting simultaneously on a body at an angle ‘ θ ’. Resultant $\vec{R}$ makes an angle α with $\vec{A}$.
 - a) Write the expression for magnitude of resultant (R). Also discuss the conditions of maximum and minimum values of resultant. [2]
 - b) A person throws a ball due east at 4m/s. Wind is blowing along north at 3m/s. Calculate the magnitude and direction of resultant vector. [2]
 - c) Define null vector. [1]
14. a) State and prove principle of conservation of linear momentum. [1+2]
 b) Gravel is dropped onto a conveyer belt at the rate of 0.5 Kg/s. Calculate extra force required to keep the belt moving at constant speed of 2 m/s. [2]
15. “The tendency of solid to change its shape, size and volume in the response of heat is termed as its thermal expansion”. Answer the following questions on the basis of given statement.
 - a) Define linear expansivity of solid materials. [1]
 - b) Show that $\alpha = \frac{\beta}{2}$, where the symbols carry usual meanings. [2]
 - c) An aluminum rod when measured by steel scale, both being at 25°C appears to be 1m long. If the scale is correct at 0°C . Calculate (i) length of scale at 25°C (ii) length of rod at 0°C . [$\alpha_{Al} = 26 \times 10^{-6}/^\circ\text{C}$, $\alpha_S = 12 \times 10^{-6}/^\circ\text{C}$] [2]

16. Dulong and Petit's demonstrated an experiment to determine the absolute expansivity of liquid by considering U-shaped glass tube. On the basis of this experiment. Answer the following questions:
- State the principle of this experiment. [1]
 - Derive an expression for the absolute expansivity of liquid. [2]
 - A U- shaped glass tube is filled with an experimental liquid and the limbs are adjusted at 0°C and 100°C. The height of liquid level at one limb is 160 cm and that in hot limb is 1.8 cm higher. Calculate the apparent expansivity of liquid. [2]
17. a) Derive the mirror formula for convex mirror and show that

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R}$$
 [3]
- A mirror forms an erect image 30 cm from the object and twice the height. Where should the mirror be placed from the real object? [2]
18. a) Deduce the relation between real depth and apparent depth. [3]
- A transparent cube of 12 cm edge contains a small air bubble. Its apparent depth viewed through one face of the cube is 6 cm and when viewed through the opposite face is 2 cm. What is the actual distance of the bubble from the first face? [2]
19. a) State the Coulomb's law of electrostatics. [1]
- Write an expression for force exist between two charges, 'Q' and 'q' when they are separated by distance 'a'. Also, define one coulomb charge. [1+1]
 - Two charges having magnitude $-4\mu\text{C}$ and $+1\mu\text{C}$ are placed at distance 2 m apart. Find the position of null point. [2]

OR

- Use Gauss Theorem to derive an expression for electric field intensity due to infinite line charged conductor. [3]
- An electron of charge $1.6 \times 10^{-19} \text{ C}$ is situated in a uniform electric field and intensity 120,000 V/m. Find the force and acceleration on it. [2]

Group – C

Give long answer to the following questions.

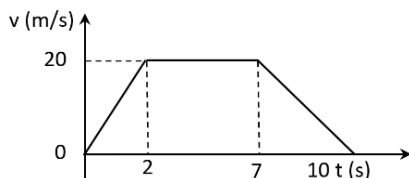
[3 × 8 = 24]

20. a) Define projectile motion. [1]
- Find the expression of horizontal range and maximum height for a projectile projected at an angle 'θ' from ground. [3]
 - A body is projected upwards making an angle θ with the horizontal with a velocity of 300 m/s. Find the value of θ so that the horizontal range will be maximum. Hence, find its range and time of flight. [3]
 - Find the angle of projection at which horizontal range and maximum height of a projectile are equal. [1]

OR

- Using velocity-time graph, derive the expression $v^2 = u^2 + 2as$, where the symbols carry their usual meaning. [2]

b) From the v-t graph:



i) Find the acceleration from 0 to 2 sec. [1]

ii) Find the displacement from 0 to 10 sec. [1]

c) An object is dropped from the top of the tower of height 156.8 m. At the same time, another object is thrown vertically upward with the velocity of 78.1 ms^{-1} from the foot of the tower, find when and where the object meet? [3]

d) Can a body have zero velocity but finite acceleration? Explain. [1]

21. A technician measures the specific heat capacity of unknown solid by using the method of mixture.

a) Draw proper calorimeter arrangement and state the principle used. [1]

b) Express the mathematical relation for the heat loss and heat gain by corresponding bodies. [2]

c) Based on the expression of heat exchange, deduce equation of specific heat capacity of solid. [1]

d) Water equivalent of a body is same as thermal capacity in CGS system. Justify. [2]

e) A copper pot with mass 0.5 kg contains 0.170 kg of water at a temperature of 20°C . A 0.25 kg block of iron at 85°C is dropped into the pot. Find the final temperature of mixture assuming no heat loss to the surrounding. [Specific heat capacity of Copper and Iron = $0.093 \text{ cal gm}^{-1}\text{C}^{-1}$ and $0.1 \text{ cal gm}^{-1}\text{C}^{-1}$ respectively.] [2]

22. a) Explain how can you charge a body positively by the method of electrostatic induction. [3]

b) A glass rod is rubbed with silk cloth. The glass rod acquires a charge of $+19.2 \times 10^{-19}\text{C}$:

i. Find the number of electrons lost by glass rod. [1]

ii. How much mass is transfer from glass to silk? ($m_e = 9.1 \times 10^{-31}\text{kg}$) [1]

c) ABC is an equilateral Δ of side 5 cm. The charge of $+5 \mu\text{C}$ and $+10 \mu\text{C}$ are placed at A and B. Calculate the magnitude and direction of electric field intensity at C. ($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$) [3]

☞ - The End - ☞